Pay the city or fix the building?

DOMAIN PROBLEM

How can Boston office building owners minimize the cost of meeting BERDO emissions requirements?

DATA SCIENCE PROBLEM

How can we estimate the impact of retrofits on real buildings when their post-retrofit outcomes are unknown?

Konstantinos Theodoropoulos · Zachary Wegrzyniak · Kateryna Lysytsyna

github.com/wegry/DS5500

99 of 297 Boston offices must pay or fix by 2030

Pay

An **Alternative Compliance Payment (ACP)**: **$234 per ton** of CO_2 over the limit, every year [4].

Cheap now. Grows as the limit drops every five years.

VS.

Fix

	Retrofit
Single	LED · wall insulation · roof insulation · windows
Pkg 1	Envelope: walls + roof + windows
Pkg 2, 4	LED + heat pumps
Pkg 3	Everything

Table 1. The eight retrofits compared.

Fixing costs more up front and saves every year after. **So the answer depends on how far ahead you look.**

24 Farnsworth St: a real Boston office we can observe



Fig.2 Photo: Boston Office Spaces [8]

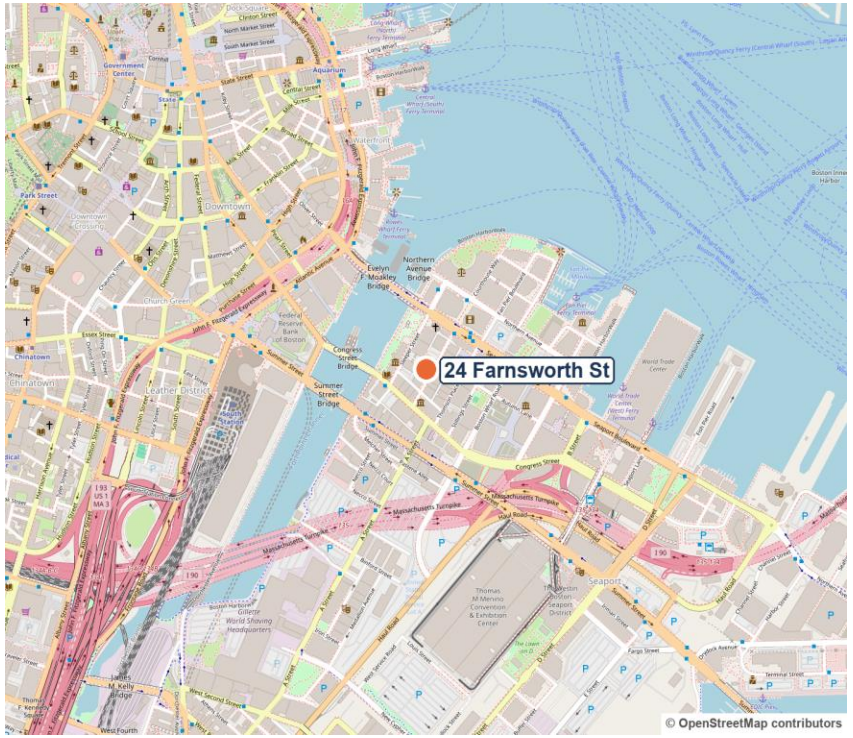


Fig.2 Fort point, accross the chanel from downtown [8]

BUILDING FACTS

Built	1915
Last renovated	1989
Floor area	79,000 ft² [1]
Natural gas	65% of energy
Current emissions	**4.8 kg CO_2e/ft²/yr**

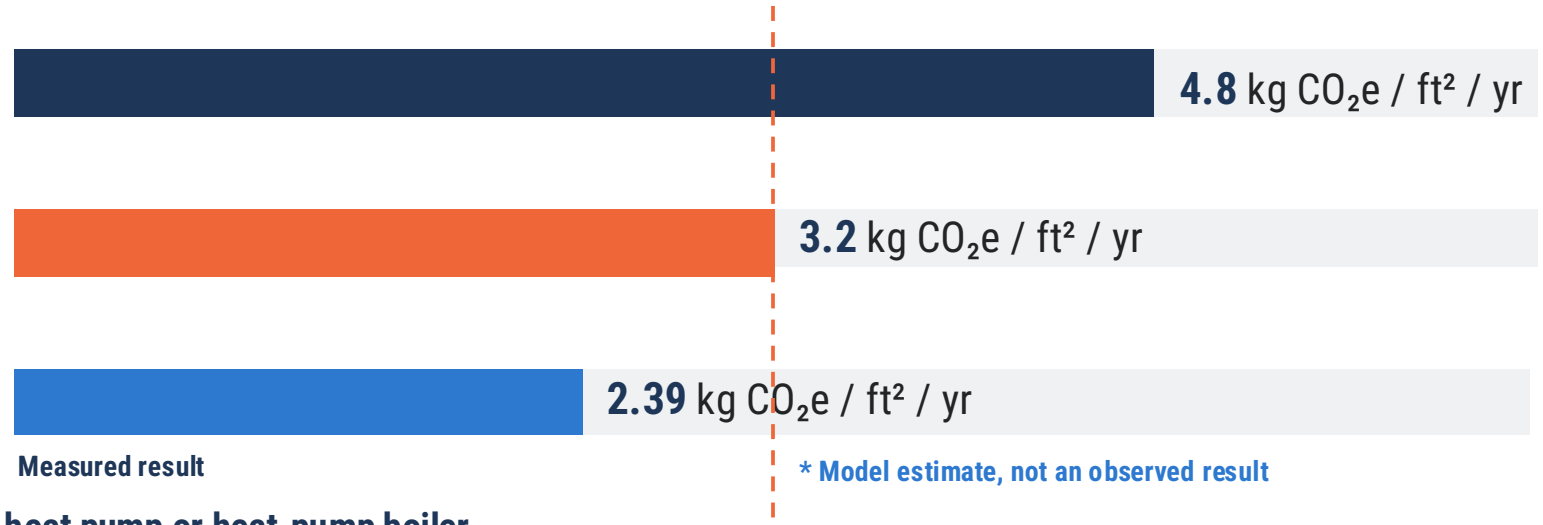
Source: Boston Office Spaces building photo; OpenStreetMap map; Boston BERDO disclosure.

> We have its current energy data—not its actual response to a future retrofit.

The missing data: what would this building emit after a retrofit?

Farnsworth Street

2030 office limit: 3.2 kg CO_2e / ft² / yr [2]



Package 2 = LED + variable-speed rooftop heat pump or heat-pump boiler

Our data science target is a predicted outcome: post-retrofit emissions for each building × option.

What does each dataset actually look like?

BERDO 1 real building	**Identifier**	**berdo_id, building_address**
	Matching fields	**property_type, gfa_sqft** (gross floor area), **eui_kwh_sqft** (energy use intensity)
	Observed outcomes	**fuel kWh, fuel shares, ghg_kgco2e** (anual greenhouse gas emission)
ComStock 1 synthetic model × upgrade	**Identifier**	**comstock_id, upgrade_id**
	Matching fields	**comstock_type, gfa_sqft** (gross floor area), **eui_kwh_sqft** (energy use intensity)
	Simulated outcomes	**electricity_kwh, natural_gas_kwh**, etc.
BTB 1 retrofit component	**Identifier**	**btb_measure_id**
	Technology	**display_name, cost_unit, lifetime_years** (equipment lifespan)
	Cost estimates	**unit_cost_low_usd, unit_cost_mid_usd, unit_cost_high_usd**

GitHub dbt staging models for BERDO, ComStock and BTB; column names are from project SQL.

We need three different grains: real property, synthetic building × upgrade, and retrofit technology.

How we estimate an unobserved retrofit outcome

Start with a BERDO office → **Find similar ComStock models** → **Keep the 10 nearest** → **Estimate retrofit change** → **Apply change to real building**

Size, EUI, fuel mix | Same building-use type | Rank by size, EUI, fuel mix | Upgraded ÷ baseline fuel use | Project its GHG intensity

MATCHING SCORE

$$d(b,c) = |\ln(A_b/A_c)| + |\ln(EUI_b/EUI_c)| + 0.5\,\Sigma_f\,|s_{b,f} - s_{c,f}|$$

b = real building; c = simulated building; A = floor area; EUI = energy use per sq ft; s = fuel share; f = fuel type

PREDICTED GREENHOUSE GAS EMISSION

$$GHG_{post} = \Sigma_f\,(GHG_{BERDO,f} \times r_f)$$

GHG_post = predicted emissions after retrofit; GHG_BERDO,f = reported baseline emissions by fuel; r_f = post-/pre-retrofit energy ratio; f = fuel type

> Nearest-neighbor matching + fuel-specific energy ratios produce building-level estimates.

After integration: one row per building × option

1 Building → simulation

berdo_id → comstock_id

match_rank, distance

2 Simulation → effect

berdo_id + upgrade_id

fuel ratios → projected GHG

3 BTB → installed cost

upgrade_id → btb_measure_id

quantity × USD per unit

FINAL MART: berdo_decision_options (schema illustration)

Retrofit option	GHG kg/ft²/yr	Meets limit?	Install	ACP (5 yr)	Energy (5 yr)	Total (5 yr)

The output is not a simple 3-table ID join: simulation effects are transferred by matching, costs by component mapping.

Are compliant retrofit choices stable when k changes?

Share of lowest-cost compliant recommendations differing from k = 10

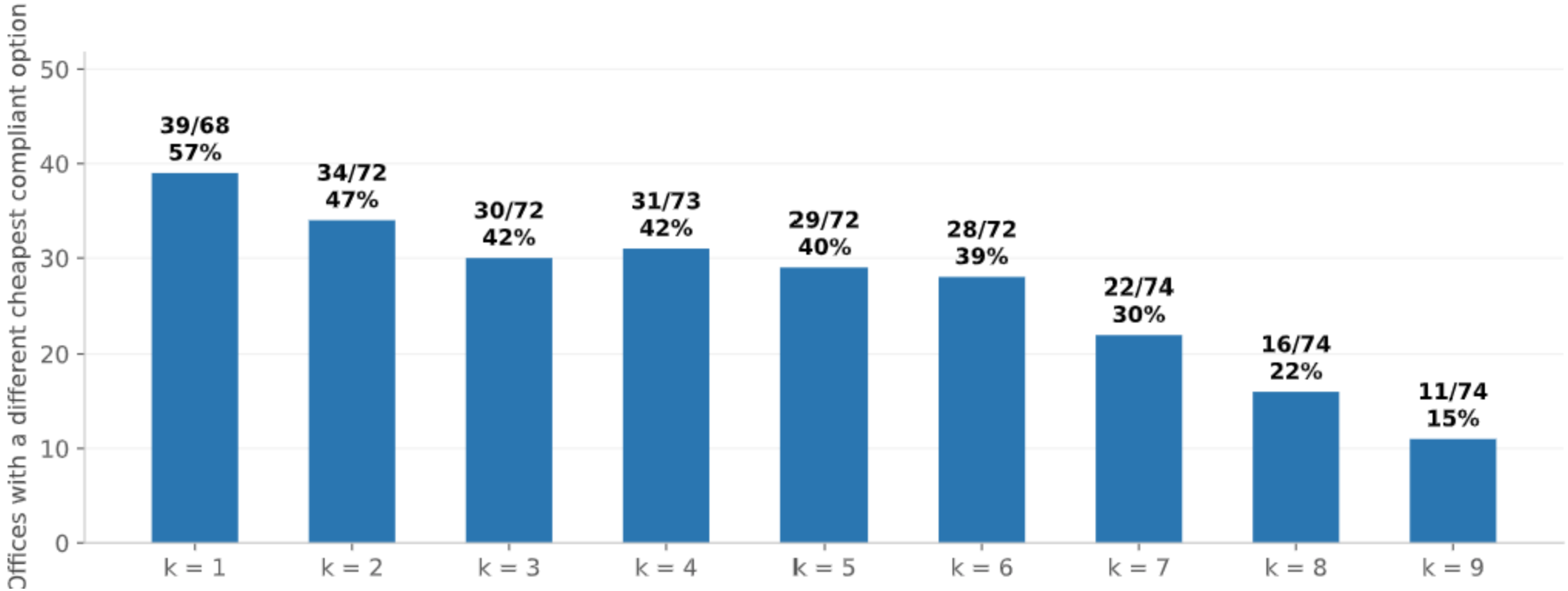


Figure 3. Sensitivity of the lowest-cost compliant retrofit recommendations to the number of nearest ComStock neighbors (k), measured as the percentage of Boston office buildings with recommendations differing from the k = 10 reference.

The current results still show meaningful choice sensitivity, even at nine neighbors.

How does the decision model choose a retrofit?

FOR EACH MATCHED OFFICE No upgrade + 4 individual measures + 4 packages

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Annual energy cost Σ_f (BERDO kWh_f × ComStock ratio_f × fuel price_f)

f = fuel type; BERDO kWh_f = reported energy consumption by fuel; ComStock ratio_f = post-/pre-retrofit energy ratio; fuel price_f = cost per kWh for each fuel.

Five-year total cost $C_{total} = C_{install} + 5 \times (C_{energy/year} + C_{ACP/year})$

C_total = total five-year cost; C_install = one-time installation cost; C_energy/year = annual energy cost; C_ACP/year = annual Alternative Compliance Payment; 5 = evaluation period in years

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Two different decisions

Cheapest overall may still exceed the limit

Cheapest compliant must have projected GHG ≤ 3.2

> It ranks nine discrete options per building.

Actual output: lowest-cost compliant retrofit at k = 10

Retrofit option	GHG kg/ft²/yr	Meets limit?	Install	ACP (5 yr)	Energy (5 yr)	Total (5 yr)
No upgrade (ACP only)	**4.80**	**No**	**$0**	**$147,776**	**$1,243,533**	**$1,391,309**
Wall insulation	4.52	No	$75,248	$122,058	$1,203,701	$1,401,007
Roof insulation	4.65	No	$62,615	$133,662	$1,222,036	$1,418,313
LED lighting	4.80	No	$96,364	$147,934	$1,208,209	$1,452,507
Pkg 2: LED + variable-speed HP	**2.39**	**Yes**	**$703,278**	**$0**	**$1,110,282**	**$1,813,560**
Pkg 4: LED + standard HP	**2.39**	**Yes**	**$703,278**	**$0**	**$1,110,282**	**$1,813,560**
New windows	4.52	No	$1,432,166	$121,929	$1,193,767	$2,747,862
Pkg 1: walls + roof + windows	4.10	No	$1,570,029	$82,892	$1,134,781	$2,787,702
Pkg 3: Pkg 1 + Pkg 2	2.17	Yes	$2,273,306	$0	$1,019,520	$3,292,826

Cheapest overall: $1,391,309 | Cheapest compliant: $1,813,560 | Compliance premium: $422,251

Table 2. Comparison of modeled retrofit options for 24–32 Farnsworth Street (BERDO ID 106476), showing projected emissions, compliance status, and five-year costs (2030–2034). The lowest-cost overall and lowest-cost compliant options are highlighted.

The cheapest option changes with the time horizon

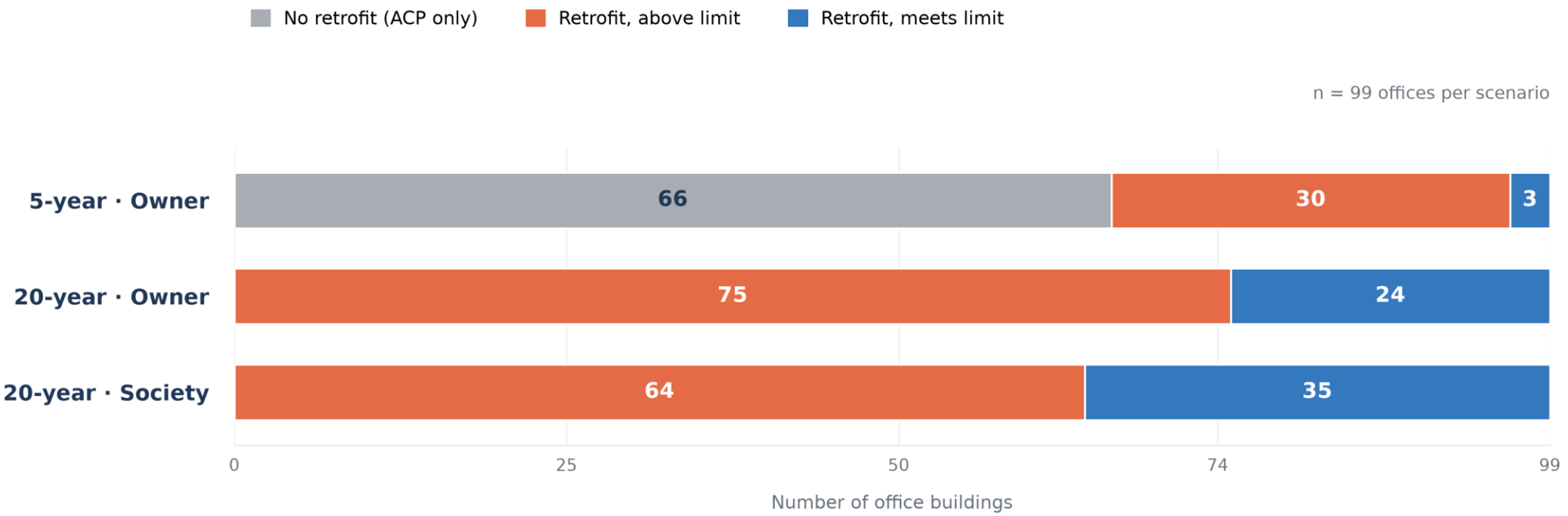


Figure 4. Distribution of minimum-cost retrofit decisions across 99 Boston office buildings exceeding the 2030 BERDO emissions limit, comparing five-year owner costs with 20-year financial and societal perspectives.

> Even over 20 years, the owner's cheapest retrofit often does not meet the emissions standard.

Back to Farnsworth: the cheapest choice need not comply



SCENARIO / LOWEST-COST CHOICE	Wall insulation	Package 2
5-year owner Selected: No retrofit	+$10k	+$422k
20-year owner Selected: Wall insulation	**−$79k**	+$113k
20-year society Selected: Package 2	−$178k	**−$303k**

Positive = additional cost; negative = savings relative to no retrofit under the same scenario.

Figure 5. Modeled cost differences for wall insulation and Package 2 relative to no retrofit at 24–32 Farnsworth Street, across five-year owner, 20-year owner, and 20-year societal perspectives.

The owner's cost-minimizing option and society's preferred option can differ for one building.

Today's ACP (Alternative Compliance Payment) rarely makes compliance the cheapest choice

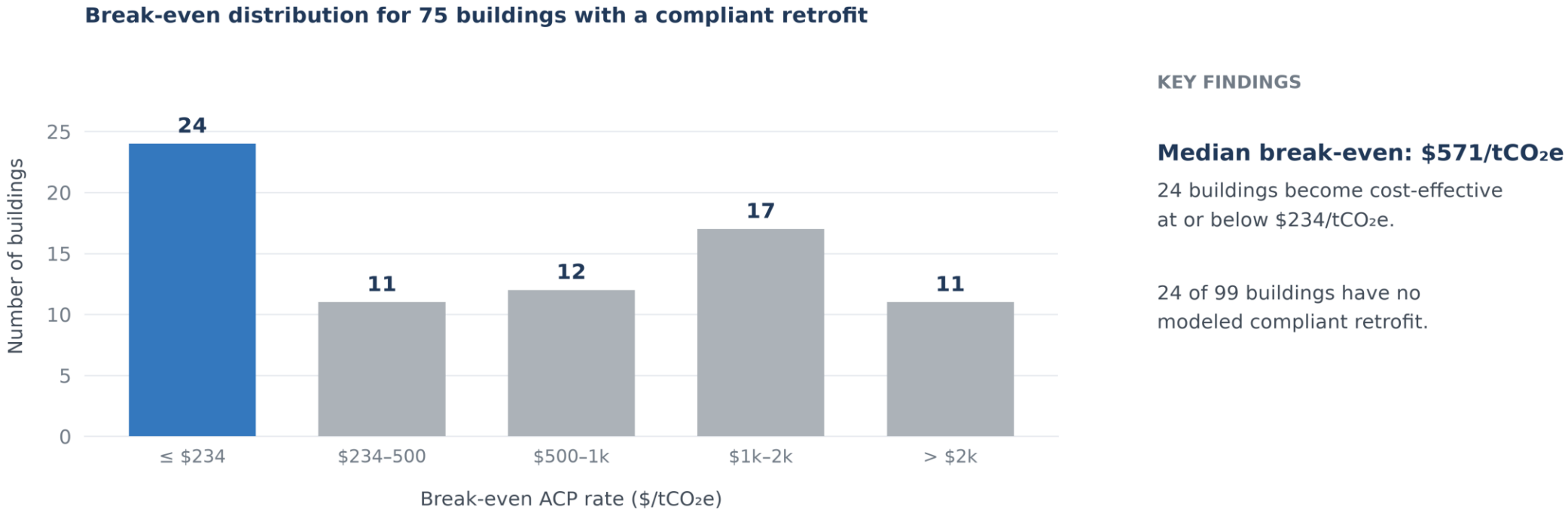


Figure 6. Distribution of break-even Alternative Compliance Payment (ACP) rates at which compliant retrofits become the lowest-cost 20-year option for building owners, across 75 Boston office buildings.

Next steps: improve the data science, then optimize the portfolio

1	**Validate predictions**	Compare estimated retrofit GHG with real post-upgrade or engineering data.
2	**Expand coverage**	Resolve missing BERDO fields; extend BTB mappings and test more upgrades.
3	**Model time explicitly**	Change electricity emissions, fuel prices and equipment costs by year.
4	**Add budget constraints**	Allocate limited capital across buildings/years with mixed-integer choices.

> Our first priority is confidence in predicted outcomes, not adding more financial complexity.

References and data credits

Data and methods

[1] City of Boston

BERDO disclosure dataset, September 2026

[2] City of Boston

BERDO emissions standards and ACP guidance

[3] National Laboratory of the Rockies

ComStock 2025 Release 3, Suffolk County

[4] NLR / Moore et al.

2024 Buildings Technology Baseline (BTB)

Scenario inputs and visuals

[5] U.S. Energy Information Administration

Annual Energy Outlook (energy-price scenarios)

[6] U.S. Environmental Protection Agency

Social Cost of Greenhouse Gases (2023)

[7] National Laboratory of the Rockies

Cambium grid-emissions scenarios

[8] Photo and map

Boston Office Spaces; OpenStreetMap contributors

github.com/wegry/DS5500/tree/main/dbt